KTA

Academic Research Organization (ARO)

A regional research support function. KTA supports researchers and clinicians throughout the entire study process—with advice, training, and quality support in collaboration with healthcare and academia.

Forum Stockholm-Gotland

Kliniska Studier Sverige

Our Services

Project Management
Coordinate all aspects of the study and ensure compliance with regulatory requirements

Consultations
Provide free-of-charge advice on study design, regulatory requirements, applications, and appropriate referrals.

Revise Study documentation
Review the study protocol and participant information

Trainings
Good Clinical Practice
PI-school
Research nurses
Research support staff

eCRF
Develop or review eCRFs in REDCap and Greenlight Guru.

TMF/ISF
Set up the investigator site file, sponsor file, and essential documents.

Applications
Prepare or review applications for submission to regulatory authorities.

Monitoring
Monitor clinical trials from initiation visits through to close-out

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The Study Protocol and Key Regulatory Aspects of Clinical Studies

Eva Löfkvist, KTA

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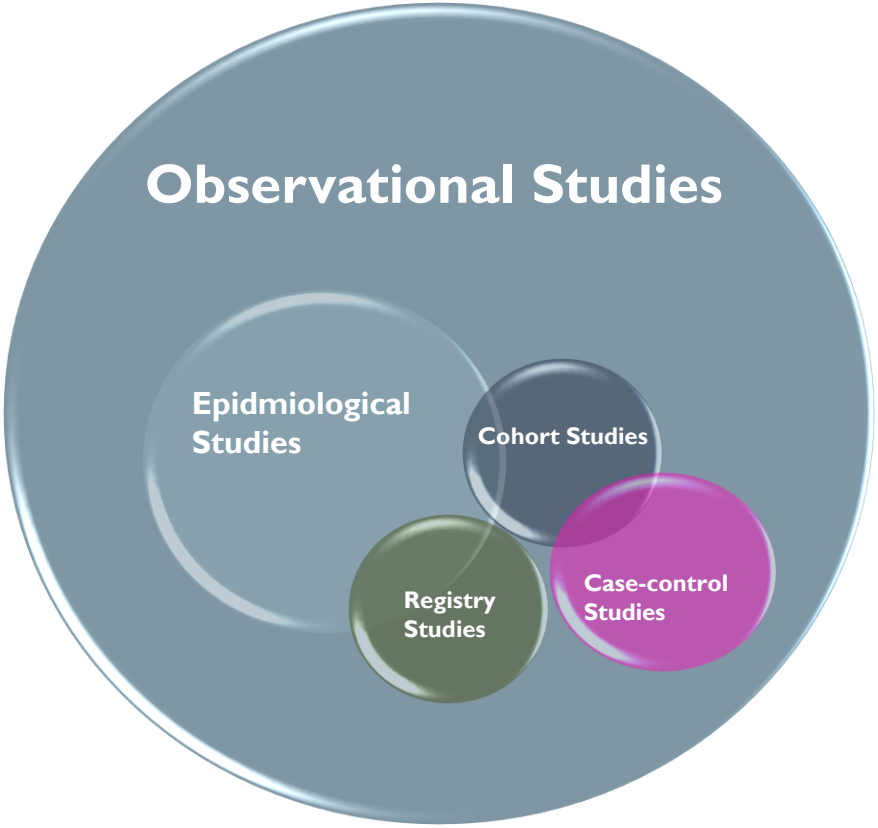
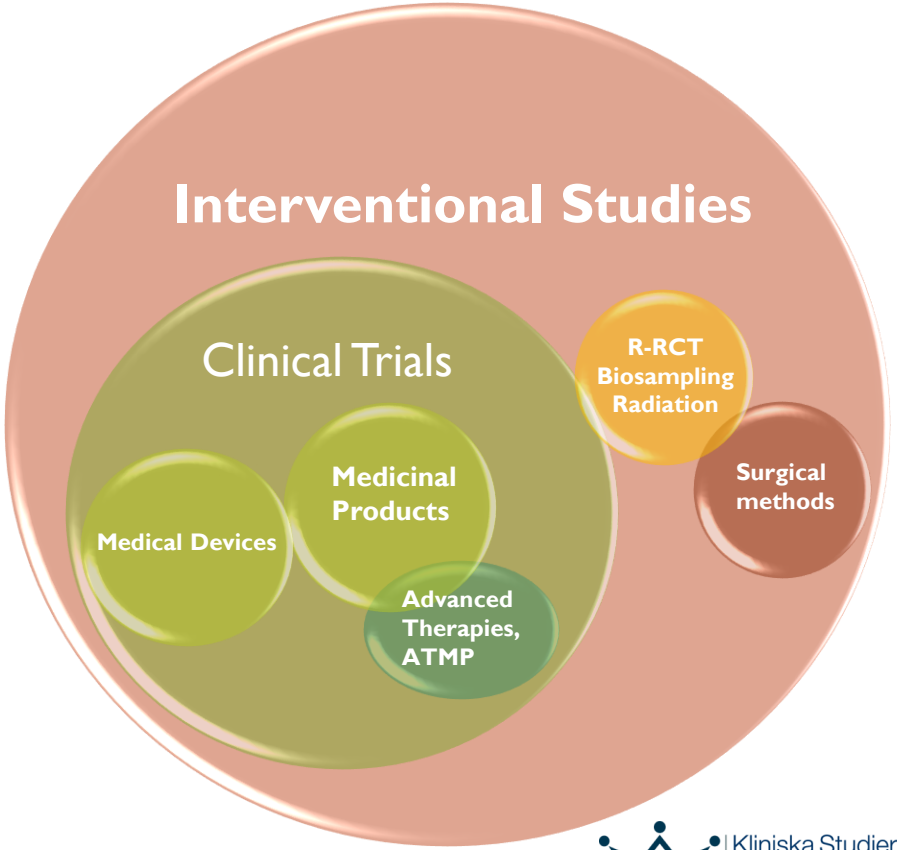
Clinical Studies

Studies on humans with medicinal or health-related objectives





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Applicable Regulatory Requirements

Clinical Studies

Helsinki Declaration
Ethics Act
General Data Protection Regulation
Archive Act*
Biobank Act*
Patient Data Act*
Patient Injury Act*
Privacy Law*

Clinical Trials

ICH E6 GCP (R3)
EU-regulation CTR
(EU Directive)
HSLF-FS 2021:109
General Data Protection Regulation
Regulation (2018:1091) Ethical reviewing
Medicines Act*

Medical Device Trials

ISO standard 14155: GCP
EU-regulation MDR 2017/746 on Medical Devices
EU-regulation 2017/748 on IVDR

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Stakeholders

Sponsor: *The organization or individual responsible for initiation, finncing and overseeing the clinical trial.*

Investigator: *The person responsible for conducting the clinical trial at a site and ensuring it follows the protocol and regulations.*

Research Principal (forskningshuvudman): *The legal entity with overall responsibility for the research being conducted (e.g., region or university)*

Authorized Representative (behörig företrädare): *Function who signs agreements and formally represents the Research Principal.*

Researcher (ansvarig forskare): *Principal researcher ensuring the study is conducted ethically and according to applicable laws and approvals.*

Participant (deltagaren): *The individual who takes part in a clinical study*

Different studies, different templates

Clinical Trial on Medicinal Products

Clinical Trial Protocol

Clinical Trial on Medical Devices

Clinical Investigation Plan, CIP

Clinical Study

Adjust template for clinical trial protocol an

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Application pathways differ

- Clinical Trial MP - CTIS
- Clinical Studies - Ethix
- Medical Device Trial - EUDAMED (under development)

The study type determines the applicable regulations!

Why is a well-written protocol critical?

A well-written protocol is critical for:

- Participant safety
- Data quality
- Feasibility
- Regulatory compliance

The protocol is used in multiple contexts

Authorities
Application Process

Site
Feasibility assessment

Research team
Clinical trial conduct

Keep in mind:

- Function
- The reader

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Key Factors of a Strong Protocol

- Engage relevant stakeholders early - pharmacy, clinicians, statisticians, etc.
- Ensure alignment between objectives, endpoints, and analysis
- Adapt the level of detail
- Involve patients
- Ensure version control – the protocol is a living document
- Use templates and checklists

The Protocol Structure

- Synopsis
- Background
- Objectives and endpoints
- Benefit-risk
- Trial design
- Subject selection
- Screening
- Informed consent procedure
- Trial treatments
- Methods for measurement of endpoints
- Handling of adverse events
- Statistics
- Quality
- Ethics
- Substantial changes to the trial
- Data governance
- Notifications

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Background and Objectives

What Do We Aim to Achieve?

Background – why the study is needed and how it builds on existing knowledge

Primary objective = the trial's main research question

Secondary objectives: complementary questions, e.g. quality of life, safety, subgroups

Considerations:

- Avoid vague wording like "evaluate the effect" - specify what and when
- Each question should have a clearly defined endpoint
- A clear objective leads to a clear study design

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Endpoints – how the study is measured

- **Endpoints** define how the objective is measured in practice
- **Primary endpoint:** directly linked to the primary objective
- **Secondary endpoints:** complement the overall picture and link to secondary objectives
- Only one primary endpoint – essential for study power (i.e. determining the sample size needed for statistically significant results)
- Endpoints should be measurable: objective, relevant, time-defined, and with low variability

Benefit-Risk Assessment - The Ethical Balance

- Risks related to the investigational product or medical device
- Risks associated with study procedures compared to standard clinical practice
- Avoid stating that there are no risks
- How will risks be minimized?
- What are the potential benefits for participants?
- For medicinal products: do the benefits outweigh the risks?
- For radiation: specify dose, number of procedures, and justify the risk–benefit balance
- Consider both study arms in the risk–benefit assessment (including placebo/reference arm)

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Flow Chart – a Visual Aid

Flow Chart				
Procedure	**Screening** Day/Week X Inclusion visit	**Visit 1** Baseline	**Visit 2** Day/Week X (±10 days)	**Visit 3** Day/Week X (±10 days)
Incl/exclusion criteria	✓	✓	✓	✓
Informed consent	✓	✓	✓	
Medical history/ concomitant medications	✓	✓		
Randomisation	✓*	✓*	✓	✓
Instructions for handling the medicinal product(s)	✓	✓	✓	✓
Fill out EQ-5D		✓	✓	✓
X-ray (CT)		✓	✓	✓
Adverse Events (AE & SAE)		✓	✓	✓

**Randomisation can be done at, e.g., inclusion visit or visit 1; adjust the table for the trial.*

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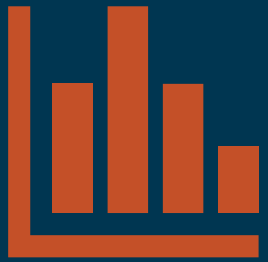

A robust study design, supported by statistical expertise, is essential to ensure **ethical use** of patients, staff, and resources.

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How is quality ensured?

Quality Assurance (proactive)

- Staff training
- Written procedures
- Identification of source data
- Risk assessment and risk management

- Identify critical quality factors
- Procedures for handling deviations

Quality Control

- Monitoring
  - Before, during and after trial
- Audits and Inspections

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Start with the End in mind

- Invest more time in planning and designing the protocol
- Consider and plan all processes
- Version handling od documents

Reduces risk of misinterpretation
- Improves data quality and conclusions
- Enhances participant safety

Reduces number of amendments = less administrative burden

Saves resources in the long term

Well prepared studies deliver robust outcomes

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Templates and KSS

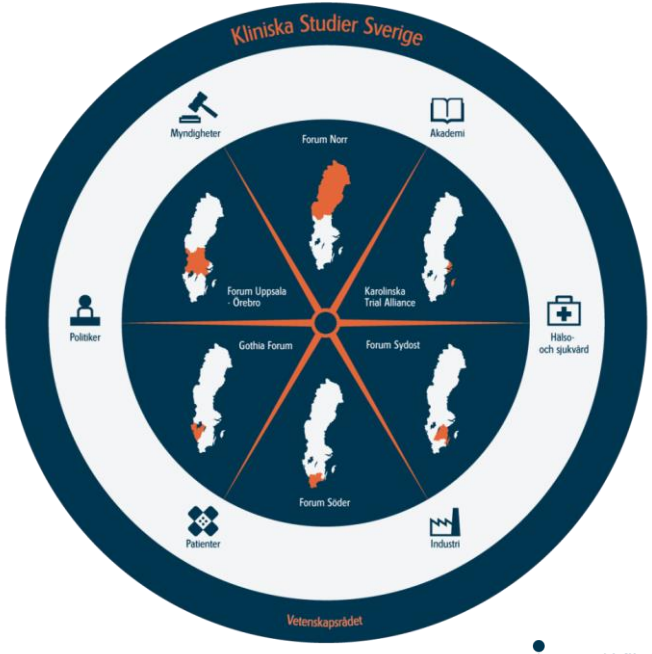
- Kliniska Studier Sverige – national collaboration within Sweden's healthcare regions
- Enhance the feasibility of conducting clinical trials in Sweden
- Feasibility Sweden

Generic templates for protocol and checklists:

KSS Templates

Generic Log-templates found here:

Apotekarsocieteten Mallar



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Thanks!

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